FLORINDOLAB · USER GUIDE

FlorindoAudioSuite

26 free audio plug-ins and a rack, in VST3 and Audio Unit formats


GAIN


WIDTH


TUNER


PEAK


PHASE


SCOPE


LOUD


REF


BPM


ParamEQ


GraphEQ


AnalogEQ


FILTER


COMP


MultiCOMP


DeESSER


GATE


LIMIT


PUNCH


ECHO


SWIRL


VERB


PITCH


DRIVE


THUMP


CAB


RACK

CONTENTS

Contents

Installation (Windows, Mac) · Common features · FlorindoRACK · Sidechain · The plug-ins, by family:

UTILITY

- **FlorindoGAIN** · Gain and polarity
- **FlorindoWIDTH** · Stereo width and monitoring
- **FlorindoTUNER** · Tuner

ANALYSIS

- **FlorindoPEAK** · Level metering
- **FlorindoPHASE** · Phase and mono compatibility
- **FlorindoSCOPE** · Spectrum analyser
- **FlorindoLOUD** · Loudness (EBU R128)
- **FlorindoREF** · Reference track comparison
- **FlorindoBPM** · Tempo, key and Tap Tempo

EQ

- **FlorindoParamEQ** · 8-band parametric EQ
- **FlorindoGraphEQ** · 10-band graphic EQ
- **FlorindoAnalogEQ** · 5-band EQ, analogue front panel
- **FlorindoFILTER** · Resonant high-pass and low-pass

DYNAMICS

- **FlorindoCOMP** · Compressor
- **FlorindoMultiCOMP** · 4-band compressor
- **FlorindoDeESSER** · De-esser
- **FlorindoGATE** · Gate and expander
- **FlorindoLIMIT** · True Peak limiter
- **FlorindoPUNCH** · Transient shaper

EFFECTS

- **FlorindoECHO** · Stereo and ping-pong delay
- **FlorindoSWIRL** · Chorus, flanger, phaser
- **FlorindoVERB** · Reverb and convolution
- **FlorindoPITCH** · Pitch shifting

CHARACTER

- **FlorindoDRIVE** · Saturation and soft clipper
- **FlorindoTHUMP** · Bass preamp
- **FlorindoCAB** · A/B cabinet IR loader

Installation on Windows

FlorindoAudioSuite runs on Windows 10 and 11 (64-bit), in any VST3 host: Reaper, Cubase, Studio One, FL Studio, Ableton Live, DaVinci Resolve (Fairlight)...

1. **Unblock the zip.** Before extracting it: right-click the zip file, Properties, tick “Unblock”, OK. Otherwise Windows may prevent the plug-ins from loading.
2. **Extract the zip** anywhere you like.
3. **Copy the .vst3 folders** from the VST3 folder to `C:\Program Files\Common Files\VST3`. Windows asks for administrator confirmation: accept. You can copy only the plug-ins you need.
4. **Open your host** and rescan for new plug-ins if needed (in Reaper: Options, Preferences, Plug-ins, VST, “Re-scan”).
5. **The plug-ins appear** as “Florindo...”, vendor FlorindoLAB, sorted by category.

WHERE ARE MY PRESETS?

The presets you save are stored in `%APPDATA%\FlorindoLAB\Audio Presets`, one folder per plug-in. To back them up or move them to another computer, simply copy this folder. FlorindoCAB's IR library (folders, favourites, recent) is in `%APPDATA%\FlorindoLAB\CAB`.

UNINSTALLING

Delete the “Florindo....vst3” folders from `C:\Program Files\Common Files\VST3`.

STANDALONE APPLICATIONS

Every plug-in also exists as a standalone application, in the Standalone folder: handy for tuning an instrument, analysing a song or measuring a sound without opening a host. Run the .exe file directly (no installation), then choose your audio interface in Options, Audio/MIDI Settings.

> **Latency.** Three plug-ins add a small latency: LIMIT (2 ms look-ahead), DRIVE and THUMP (oversampling). It is reported to the host, which compensates for it automatically: your tracks stay perfectly aligned.

Installation on Mac

The macOS version runs on Intel and Apple Silicon Macs, from macOS 10.13. It comes in two formats: **VST3** (Reaper, Cubase, Studio One, Ableton Live, DaVinci Resolve...) and **Audio Unit** (Logic Pro, GarageBand, MainStage...).

1. **Extract the zip** by double-clicking it.
2. **VST3**: copy the .vst3 folders from the VST3 folder to `Macintosh HD / Library / Audio / Plug-Ins / VST3`.
3. **Audio Unit**: copy the .component files from the AU folder to `Macintosh HD / Library / Audio / Plug-Ins / Components`.
4. **Copy the applications** from the Standalone folder wherever you like, for example to the Mac's Applications folder.
5. **Unblock them on first launch.** These free plug-ins are not notarised by Apple, so macOS may block them. Open Terminal and type:
   `xattr -dr com.apple.quarantine /Library/Audio/Plug-Ins/VST3/Florindo*.vst3`
   `xattr -dr com.apple.quarantine /Library/Audio/Plug-Ins/Components/Florindo*.component`
   For an application, right-clicking its icon, then Open and confirming, also works.
6. **Restart your host** and let it scan for new plug-ins. Logic Pro validates Audio Units at startup; if needed, run the validation again from Settings, Plug-in Manager.

For your user account only, you can also use the same paths inside your home folder.

WHERE ARE MY PRESETS?

On Mac, presets are stored in `~/Library/Application Support/FlorindoLAB/Audio Presets`, shared by the VST3, AU and standalone versions.

> **Pro Tools** uses the AAX format, which is not offered for now.

GOOD TO KNOW

Common features

All plug-ins share the same layout and the same gestures.

THE PRESET BAR

< and >	previous and next preset.
List	factory presets, then your own.
Save	menu: save as a new preset, update the current preset, rename, delete, open the presets folder.

CONTROLS

Knobs	drag up or down. Double-click: default value. You can also type a value in the box below the knob.
Bypass	disables processing, with a fade, without clicks.
Listen, Key, Solo	let you hear a specific part of the signal: the band you are adjusting, the detection signal...
Delta	lets you hear only what the plug-in removes. The best way to check that a process is not doing too much.

Listening buttons (Listen, Key, Solo, Delta) are never saved with the project: when you reopen it, you always hear the normal sound.

> **Automation.** All sound-related settings can be automated from your host, with clear names.

FlorindoRACK

Every module of the suite in a single plug-in, chained in any order you like.



HOW IT WORKS

Library	on the left, modules sorted by family. Drag one into the chain, or double-click it. A copper dot marks a module already in use.
Chain	the signal flows through the modules from top to bottom. Drag a module by its header to move it. The x button removes it.
Collapse / expand	each module's arrow, or Collapse / Expand for all.
On	enables or bypasses a module without removing it.
Presets	at the top, whole-chain presets (Podcast Voice, Gentle Mastering, Drum Bus…). Each module also keeps its own preset bar, shared with the standalone plug-in.

> **Tip.** Only the modules in the chain are processed: a rack with three modules uses no more CPU than three separate plug-ins.

TUTORIAL

Sidechain

COMP and GATE can react to another track than the one they process. Classic example: music that automatically ducks when a voice speaks.

The principle: the compressor processes the music, but it **listens** to the voice to decide when to act. Here is how to set it up in Reaper (the idea is the same in other hosts).

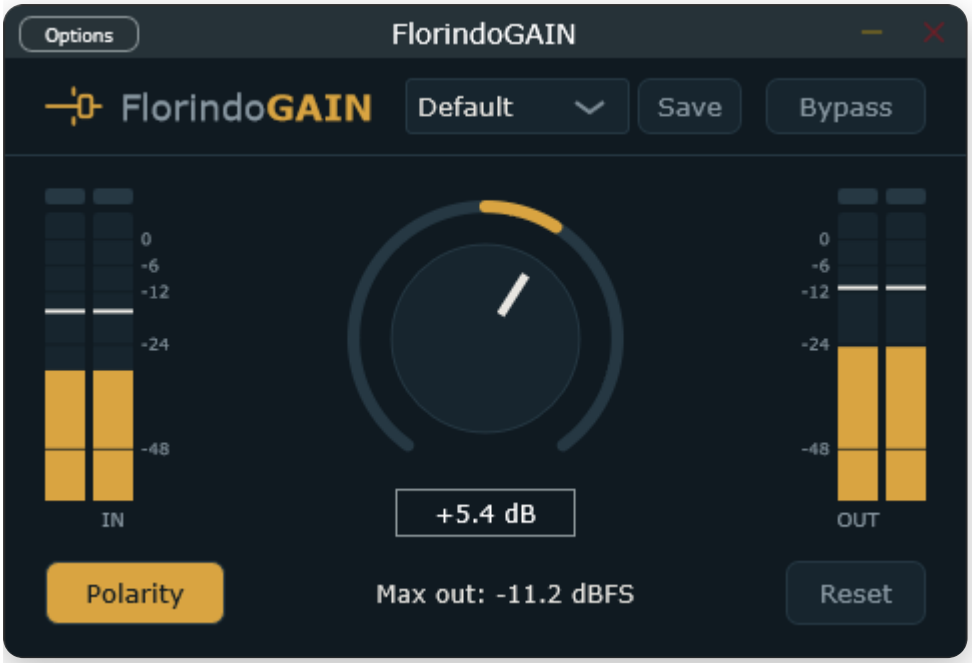
1. **Music track:** insert FlorindoCOMP, set Sidechain to **External**.
2. **Music track, Route button:** change “Track channels” from 2 to **4**. Channels 1-2 carry the music, 3-4 will receive the voice.
3. **Voice track, Route button:** add a send (“Add new send”) to the music track, with **3/4** as destination.
4. **Start playback:** when the voice speaks, the light next to the Sidechain selector turns **green**. If it stays red, the signal is not arriving: check the plug-in pin connections (pin button, top right of the plug-in window in Reaper).
5. **Adjust:** lower Threshold until the music drops by a few dB when the voice speaks, then adjust Release for a smooth recovery. The “Ducking (ext. SC)” preset is a good starting point.

With GATE, choose **Ext SC**: the gate opens in time with the sent track, for example a bass following the kick drum.

In FlorindoRACK, sidechain works the same way: the rack passes it on to the COMP and GATE modules.

FlorindoGAIN

The simplest tool in the suite: it sets the level of a track, with a clear display of input and output levels.



SETTINGS

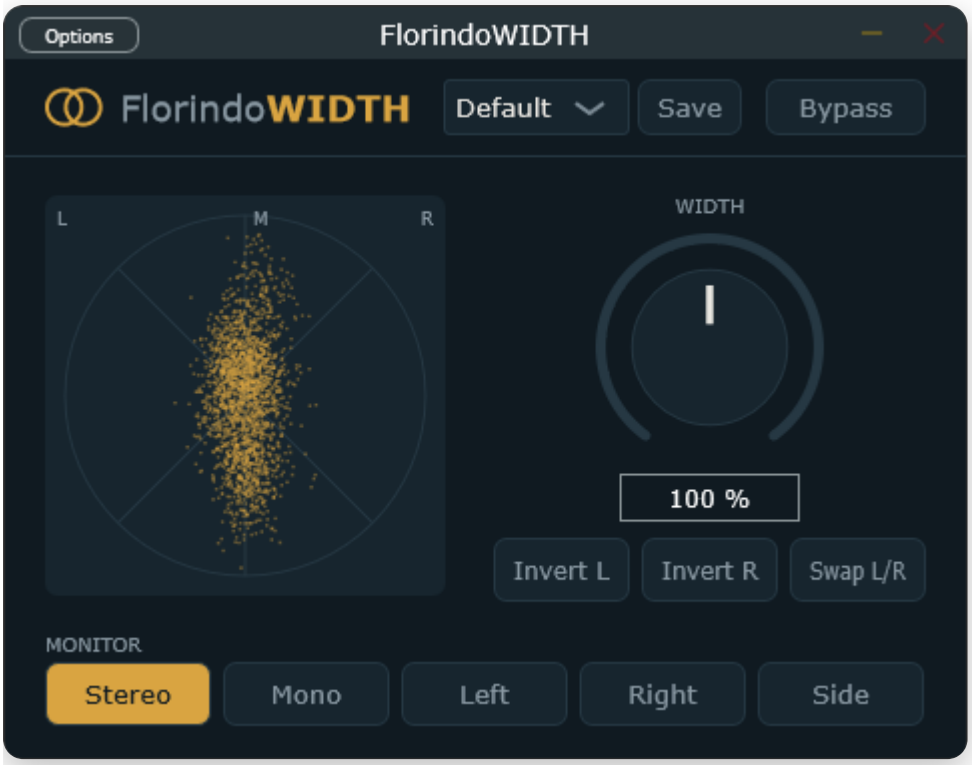
Gain	from −24 to +24 dB. Double-click to return to 0.
Polarity	inverts the signal polarity (useful to bring two microphones back in phase).
IN / OUT meters	input and output levels, with peak hold.
Max out	highest output level reached.

> **Tip.** Put GAIN at the very start of the chain to bring each track to a comfortable working level (peaks around −12 dBFS) before any other processing.

FlorindoWIDTH

Stereo width and monitoring

Widens or narrows the stereo image, and lets you listen to each part of the signal separately.



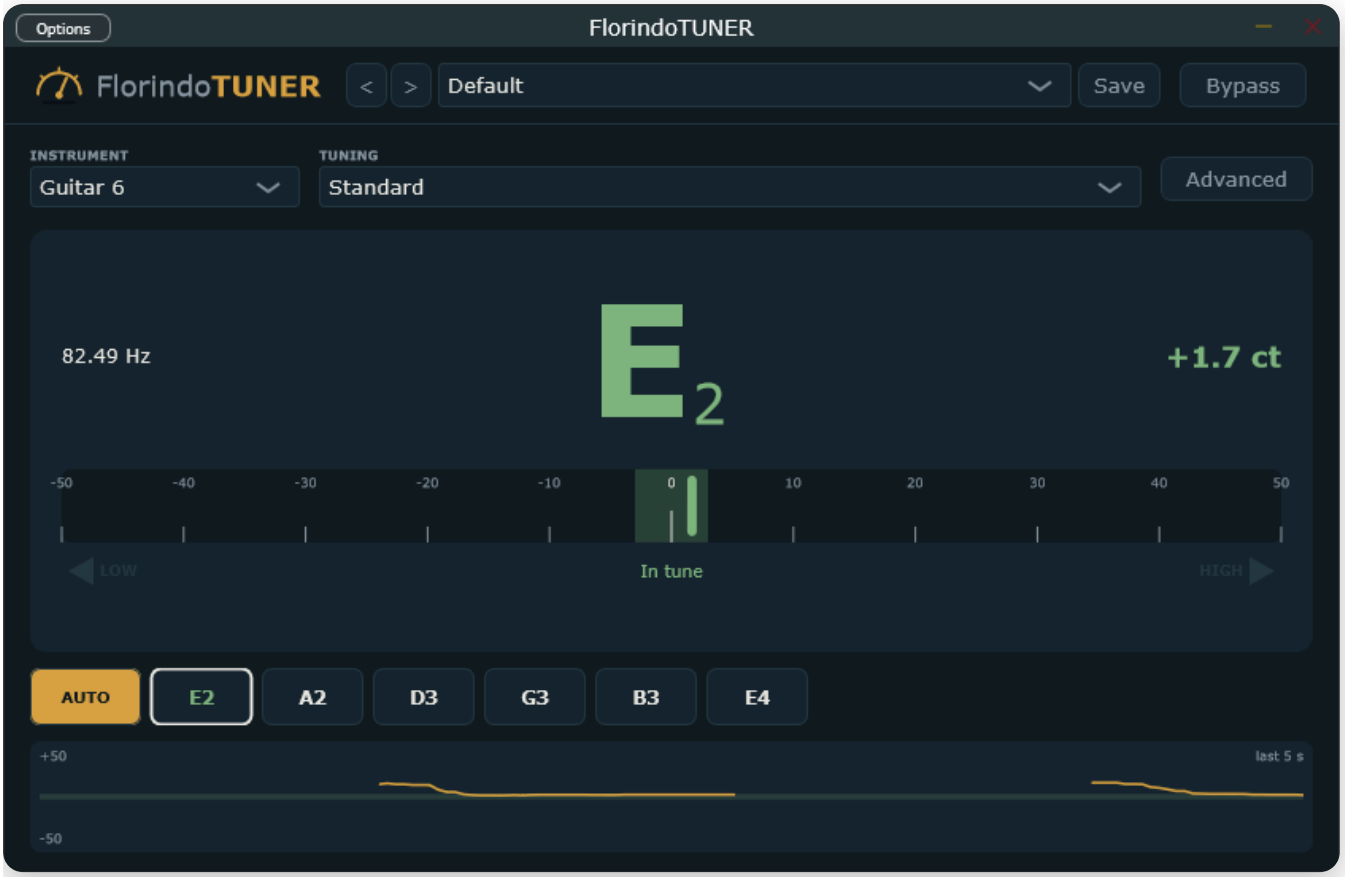
SETTINGS

Width	from 0 % (mono) to 200 % (very wide). 100 % changes nothing.
Invert L / Invert R	polarity of each channel.
Swap L/R	swaps left and right.
Monitor	listen in Stereo, Mono, Left, Right or Side (what differs between left and right).
Goniometer	shows the output stereo image.

> **Tip.** Check your mix in Mono from time to time: if an instrument disappears, it has a phase problem.

FlorindoTUNER

A chromatic and instrument tuner, accurate down to the lowest notes: ideal for down-tuned guitars and basses.

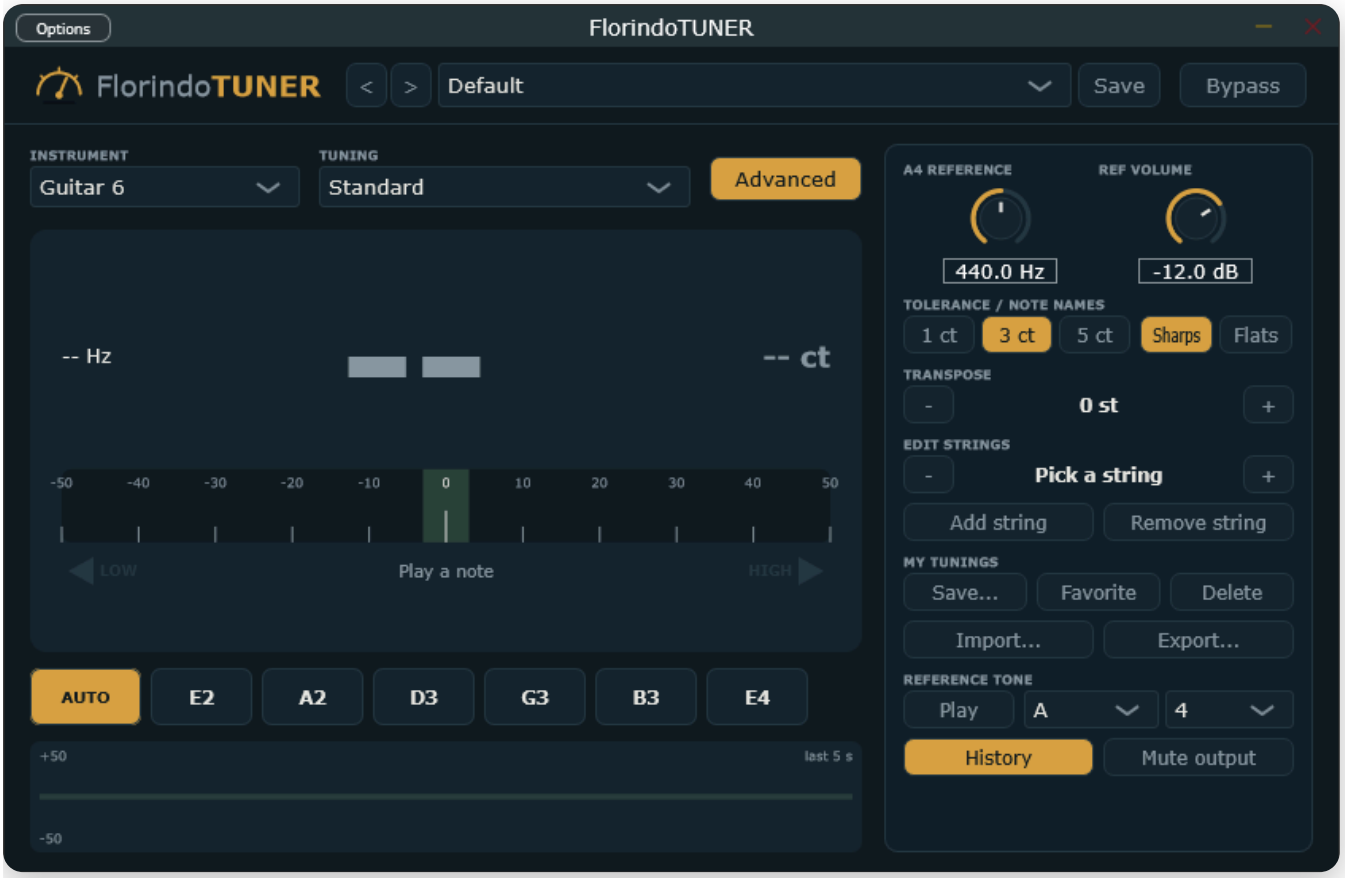


SETTINGS

Instrument / Tuning	instrument, then a preset, favourite or custom tuning. Chromatic detects any note.
String buttons	AUTO targets the nearest string; click a string to target it. A tuned string turns green.
Display	note and octave, deviation in cents, frequency, LOW / HIGH arrows. “In tune” only appears once the reading is stable.
Advanced	reference pitch (400 to 480 Hz), tolerance, sharps or flats, transposition, string editing, custom tunings, reference tone, history, silent mode.

> **Tip.** Turn on Mute output while tuning: the output is silenced, but the tuner keeps listening.

Tuning to Drop C



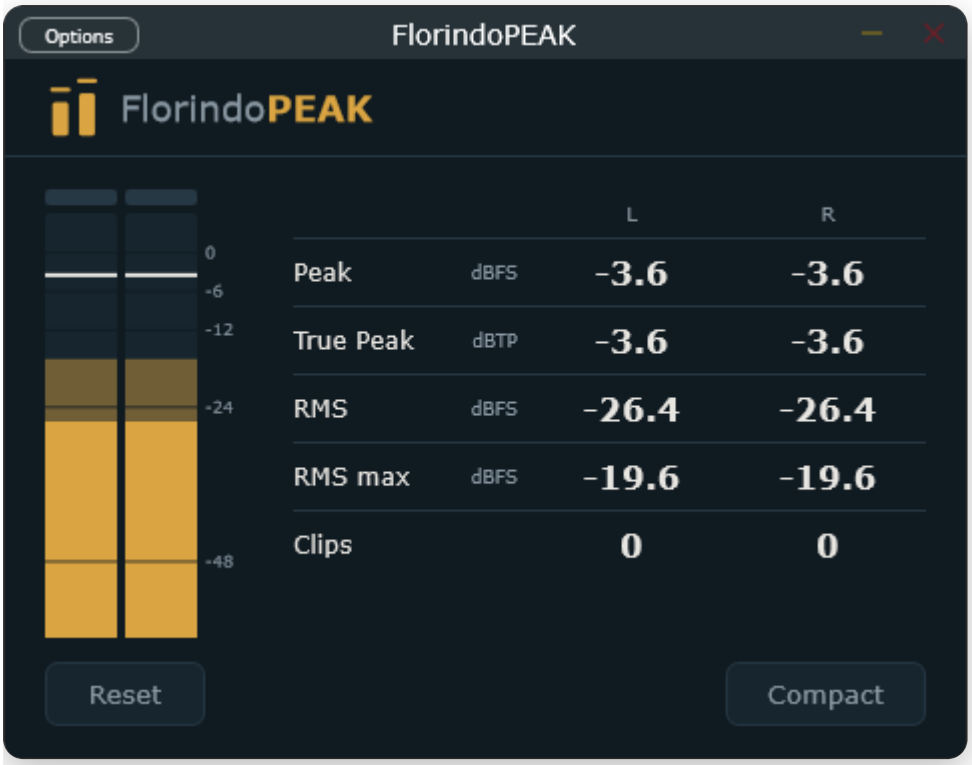
1. **Choose the instrument**: Guitar 6, then the **Drop C** tuning in the Tuning list.
2. **Play a string**: with AUTO, the tuner recognises the nearest string and outlines it. Turn the tuning peg until the needle enters the green zone and “In tune” appears.
3. **A stubborn string?** Click its button to target it directly: the tuner then only compares against that note, even if you start very far off.
4. **Your own tuning**: open Advanced, select a string and change its note with - and +. Click Save... to store it under a name, then Favorite to find it at the top of the list.
5. **A whole step down?** Transpose shifts all strings at once, from -24 to +24 semitones.
6. **Tune by ear**: Play sounds the note of the targeted string, at the chosen reference pitch.

> **Good to know.** Custom tunings are shared across all your projects. Export... saves them to a JSON file, to back them up or share them; Import... adds them to yours.

FlorindoPEAK

Level metering

An accurate level meter: peak, True Peak (peaks between samples), RMS and clip count.



SETTINGS

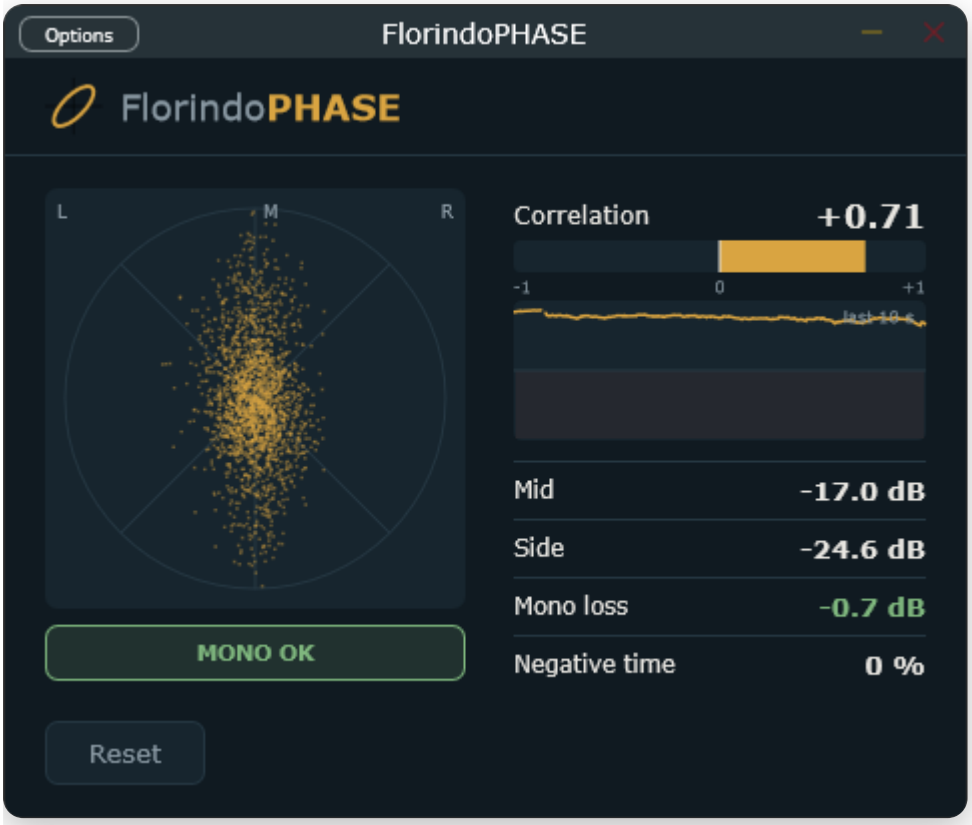
Meters	average level (RMS) and peak, with peak hold.
Peak / True Peak	highest values reached, on samples and between samples.
RMS / RMS max	average level and its maximum.
Clips	number of times 0 dBFS was exceeded.
Compact	reduces the window to the meter and the True Peak value.

> **Tip.** Before exporting for a streaming platform, make sure the True Peak stays below −1 dBTP.

FlorindoPHASE

Phase and mono compatibility

Checks the consistency between left and right, and tells you directly whether your mix survives a mono fold-down.



SETTINGS

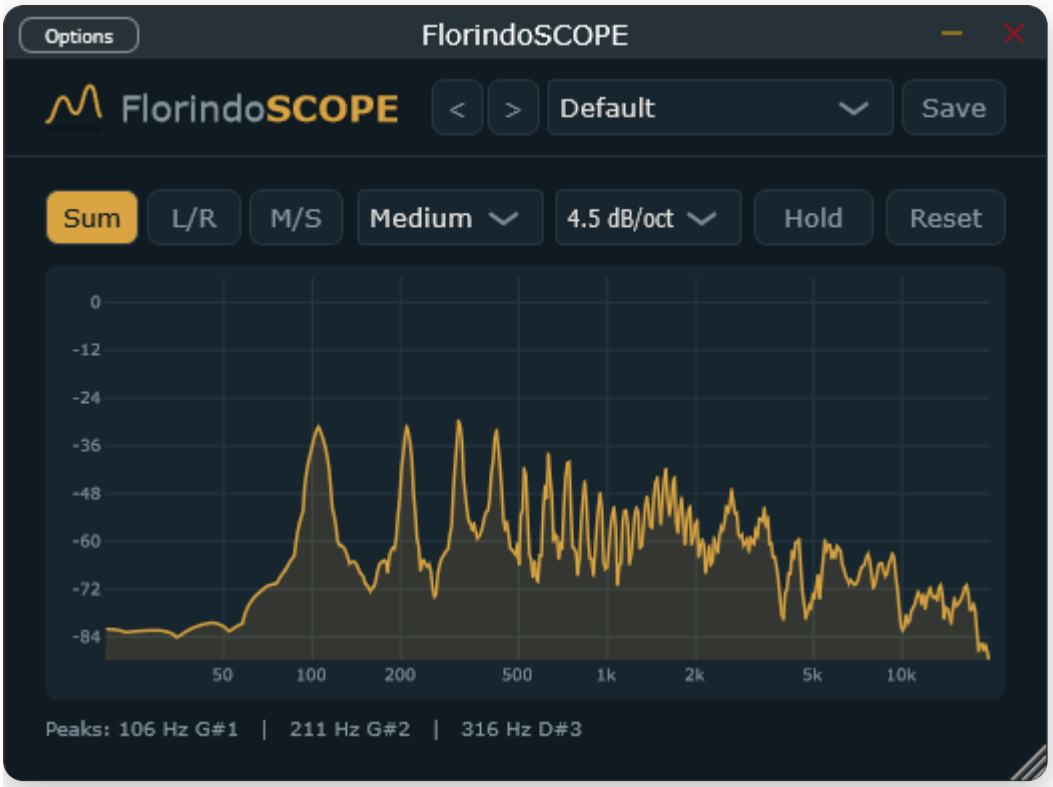
Goniometer	a vertical line = mono, a wide cloud = stereo, a horizontal line = phase problem.
Correlation	from −1 to +1. Negative = channels in opposition.
Mono OK / Check / Problem	level loss when summed to mono: under 3.5 dB is normal.
Negative time	share of time spent in negative correlation.

> **Tip.** A MONO PROBLEM warning on a stereo track often comes from a reversed microphone: try Invert L or Invert R in WIDTH.

FlorindoSCOPE

Spectrum analyser

Shows how frequencies are distributed from 20 Hz to 20 kHz, with high resolution in the low end.



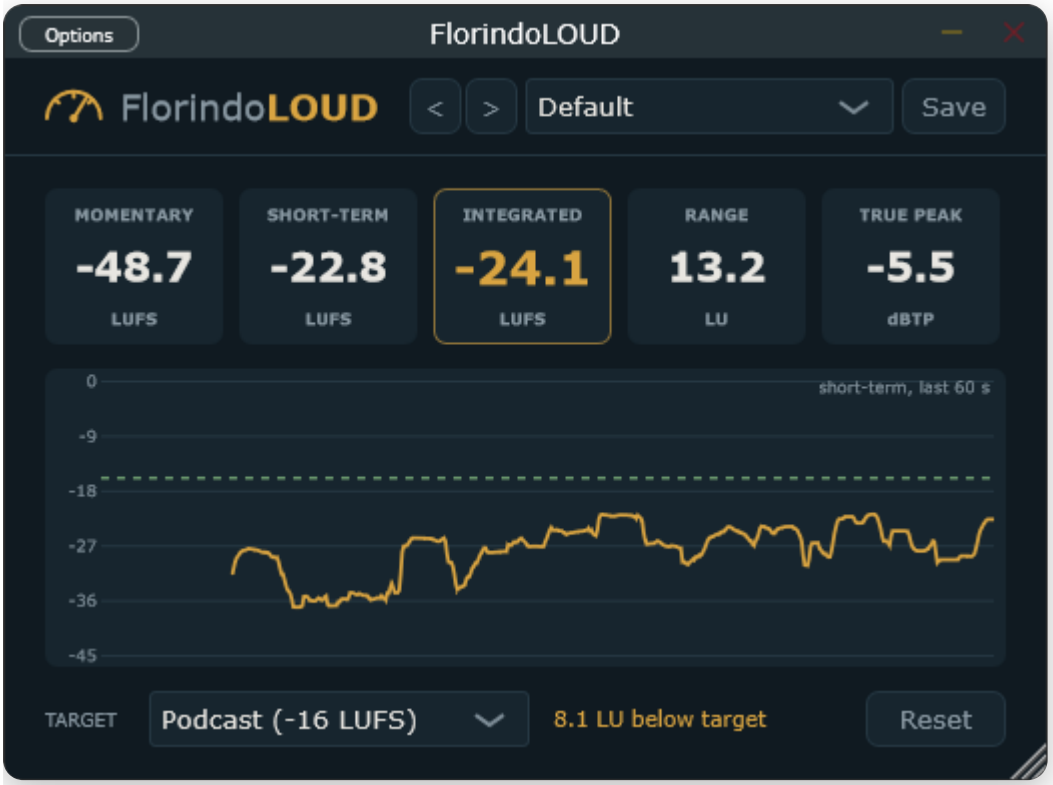
SETTINGS

Sum / L/R / M/S	spectrum of the sum, of both channels, or of Mid and Side.
Response	Fast, Medium or Slow display speed.
Slope	tilts the curve. At 4.5 dB/oct, a balanced mix looks roughly flat.
Hold	keeps the maximum curve. Reset clears it.

> **Tip.** Compare your mix with a reference track using the same slope: differences in lows or highs stand out at once. The window is resizable.

FlorindoLOUD

Measures perceived loudness according to ITU-R BS.1770 / EBU R128, the standard used by streaming platforms and broadcasters.



SETTINGS

Momentary / Short-term	loudness over 400 ms and over 3 s.
Integrated	loudness over the whole duration: the reference value.
Range (LRA)	difference between quiet and loud passages.
True Peak	highest peak between samples.
Target	value to aim for: streaming, podcast, EBU R128 broadcast...
Reset	restarts the measurement.

> **Tip.** Press Reset, play your whole song, then read the integrated value: that is the number to compare with the target.

FlorindoREF

Reference track comparison

Switches instantly between your mix and a reference track, at the same loudness, for an honest comparison.



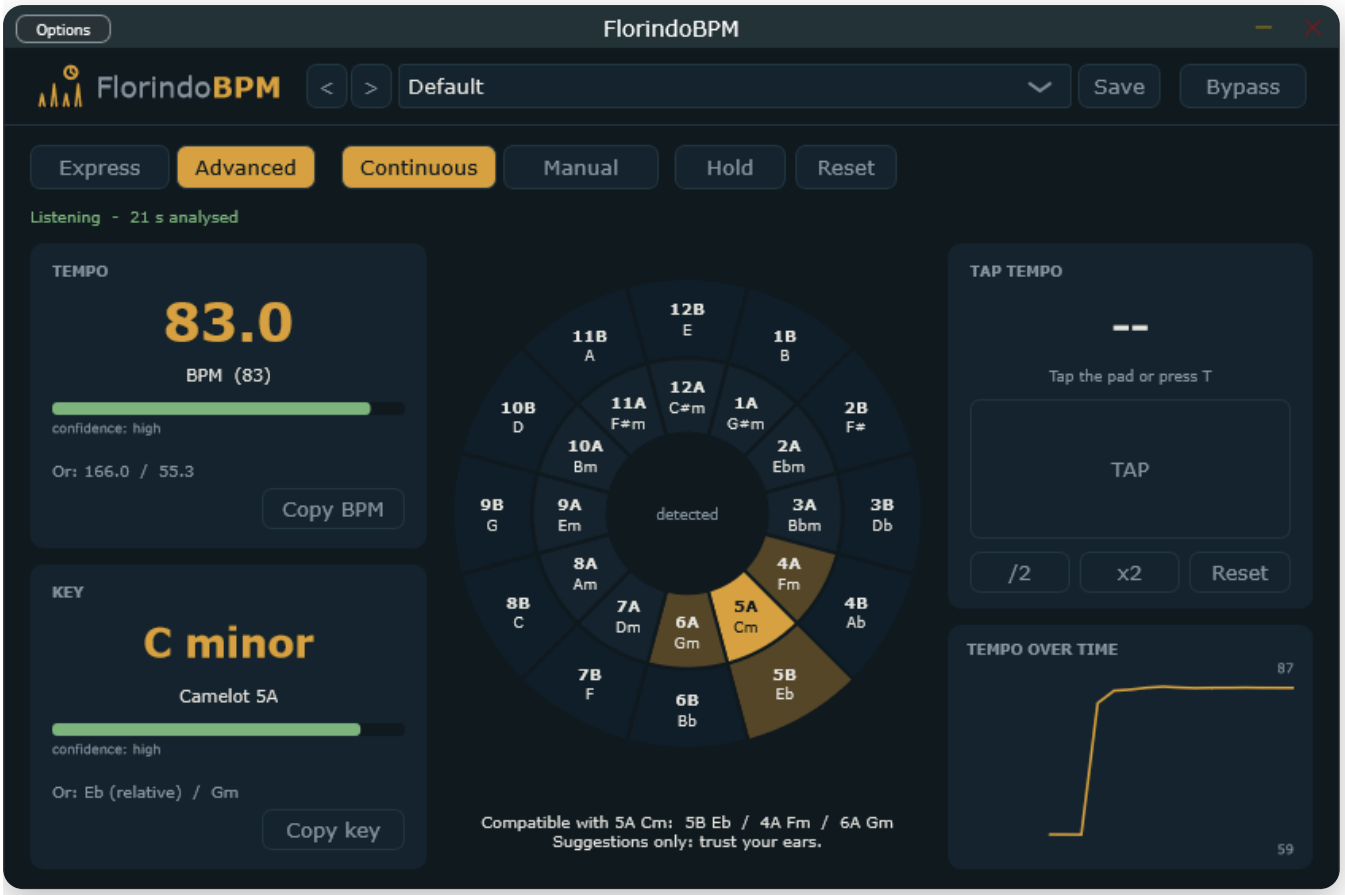
SETTINGS

Load track	loads a WAV, AIFF or FLAC file (6 minutes max).
MIX / REFERENCE	the large switch, automatable.
Match Level	matches the reference loudness to your mix. Reset LUFS restarts the measurement.
Follow DAW / Free	the reference follows project playback (with Offset), or loops on its own (click the waveform to move the playhead).
Ref Trim	manual level adjustment.

> **Tip.** Put REF last on the master bus, after the limiter.

FlorindoBPM

Finds the tempo and key of a song right inside your host, with Tap Tempo and a harmonic wheel.



SETTINGS

Tempo	detected BPM, confidence level and related interpretations (half or double tempo).
Key	key, Camelot code, confidence. The relative key is always offered as an alternative.
Copy BPM / Copy key	copy the result to the clipboard.
Tap Tempo	click TAP or press T in time with the music. /2 and x2 fix the octave; click the value to type it.
Continuous / Manual	permanent analysis (Hold freezes the results) or on demand (Start / Stop). Reset restarts the analysis.
Advanced	interactive Camelot wheel and tempo over time.

> **Tip.** Audio passes through BPM untouched: you can leave it on a track permanently.

Analysing a song and finding compatible keys



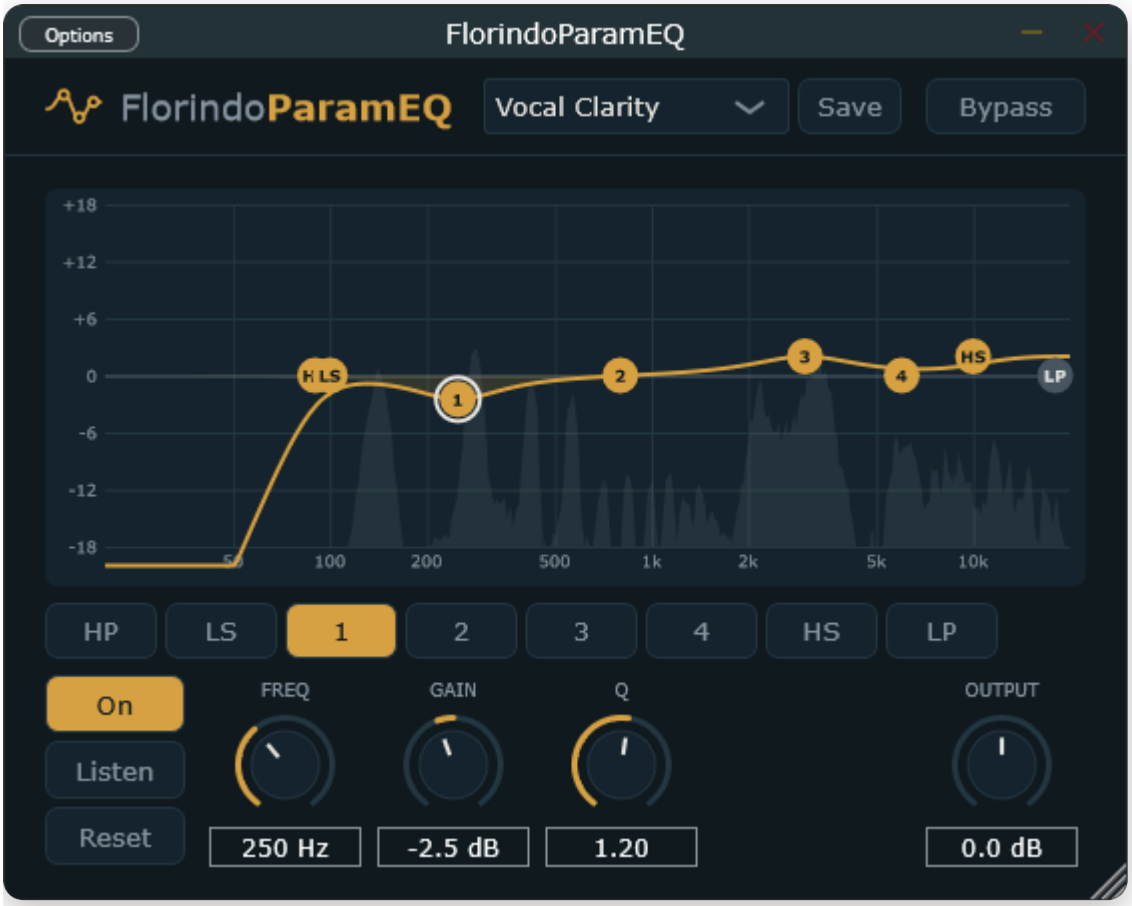
1. **Insert FlorindoBPM** on the song's track, then start playback. In Continuous mode, analysis starts by itself; the first results appear after a few seconds.
2. **Read the confidence**: green means reliable; copper or red means listen longer, or analyse a more representative passage (Manual, then Start and Stop over that passage).
3. **Half or double tempo?** If the BPM looks twice too slow or too fast, the right value is in the “Or” line. Tap Tempo settles it in a few taps.
4. **Key or relative?** A minor key and its relative major (A minor and C major) use the same notes: the relative is always offered. The mood of the song, sad or bright, helps you choose.
5. **The Camelot wheel** (Advanced mode): the detected key is in copper, compatible keys in light copper. Click a cell to explore its neighbours, and click the centre to come back.
6. **Copy the results** with Copy BPM and Copy key, for example to name your files or set the project tempo.

> **Good to know.** The wheel's compatibilities are suggestions, not rules: two songs next to each other on the wheel usually sound good together, but your ears have the final say.

FlorindoParamEQ

8-band parametric EQ

The precision equaliser: high-pass, low shelf, four bells, high shelf and low-pass, adjustable directly on the curve.



SETTINGS

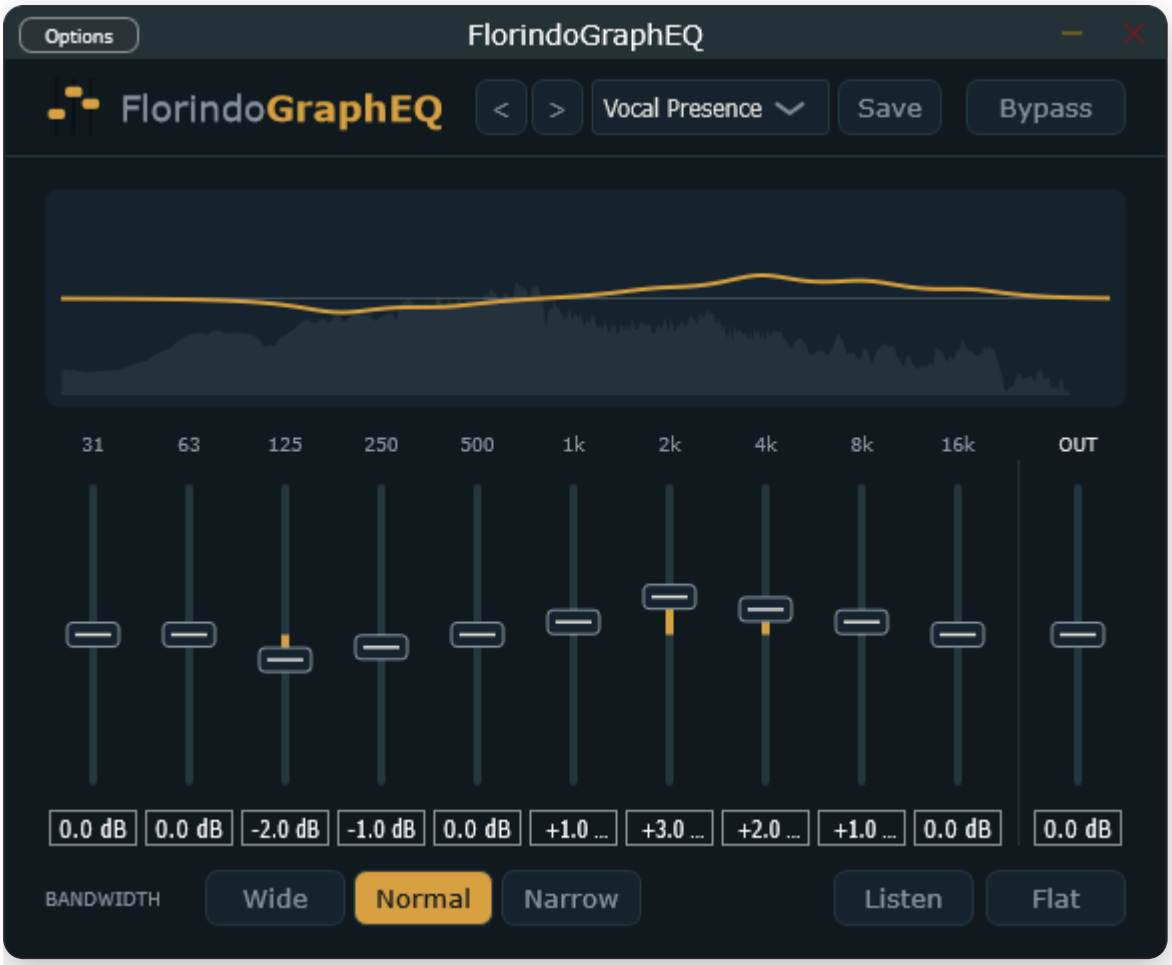
Handles	drag them on the curve to set frequency and gain.
HP / LS / 1-4 / HS / LP	selects a band and its precise settings (Freq, Gain, Q).
Listen	solos the frequency area of the selected band.
Reset	returns the selected band to its default values.
Output	output gain.

> **Tip.** To find an unpleasant frequency: bell band, high gain, Listen on, sweep until you hear it, then lower the gain.

FlorindoGraphEQ

10-band graphic EQ

Ten sliders, one per octave, from 31 Hz to 16 kHz: immediate, visual adjustments.



SETTINGS

Sliders	one per band, from 31 Hz to 16 kHz. Double-click = 0.
Bandwidth	width of the bands.
Listen	solos the band you touch.
Curve	resulting response, over the signal spectrum.

> **Tip.** Ideal for broad, quick corrections, or to tame a room's sound in a live setting.

FlorindoAnalogEQ

5-band EQ, analogue front panel

A knob-based equaliser in the style of hardware modules.



SETTINGS

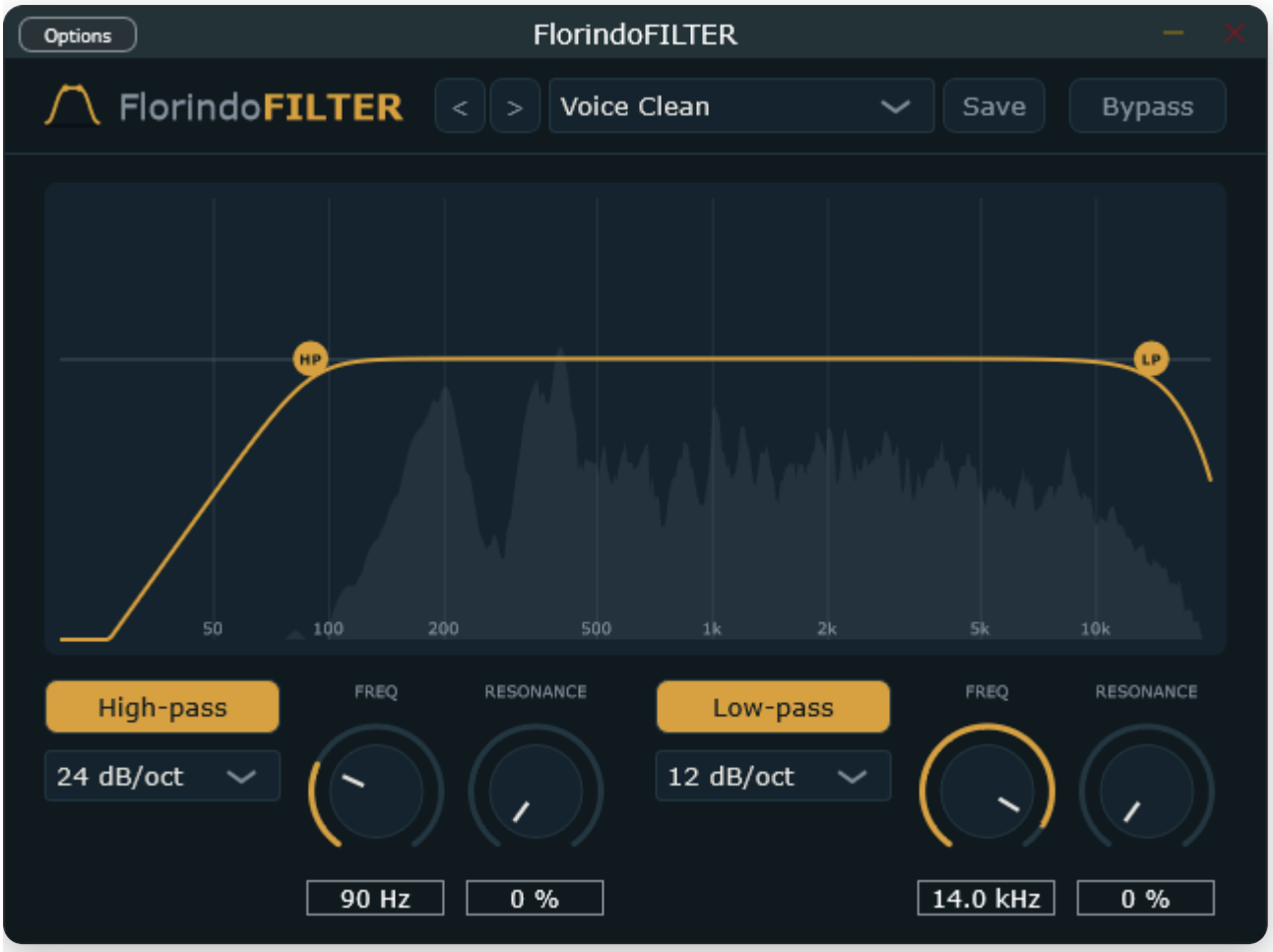
GAIN (red)	boost or cut.
FREQ (blue) and x0.1 / x1 / x10	the knob value multiplied by the chosen range.
BW (green)	band width, in octaves.
ON	enables the band.
OUTPUT, LEDs, Listen	output, meter and solo of the band you touch.

> **Tip.** Example: 300 on the x10 range = 3 kHz. A wide band gives a soft, musical colour.

FlorindoFILTER

Resonant high-pass and low-pass

Two punchy filters with adjustable slope and resonance, for cleaning up or for creative effects.



SETTINGS

High-pass / Low-pass	enables each filter.
Slope	12, 24, 36 or 48 dB/oct.
Freq / Resonance	on the curve: horizontal = frequency, vertical = resonance.

> **Tip.** Moderate resonance with an automated frequency gives the classic sweeps of electronic music.

FlorindoCOMP

Compressor

Reduces the difference between loud and quiet passages. The curve and the history make its action visible.



SETTINGS

Threshold / Ratio	threshold and amount of compression.
Attack / Release	reaction and recovery speed.
Knee	how smoothly the threshold is crossed.
Makeup / Mix	gain compensation and parallel compression.
Detector	Peak or RMS.
Sidechain	Internal or External (see the sidechain page).
Delta	lets you hear what the compression removes.

> **Tip.** With Delta, you hear at once if the attack is too fast: drum transients show up in the removed signal.

FlorindoMultiCOMP

4-band compressor

Four compressors, each on its own frequency range, with no colouration when nothing is being compressed.



SETTINGS

Crossovers	drag the vertical lines on the spectrum (120 Hz, 1 kHz, 6 kHz by default).
Per band	Threshold, Ratio, Gain, Attack, Release, On and Solo.
Mix / Output / Delta	global settings.

> **Tip.** On a master, light and slow compression of the LOW band holds the low end without squashing the rest.

FlorindoDeESSER

De-esser

Tames harsh sibilance (s, sh) in a voice.



SETTINGS

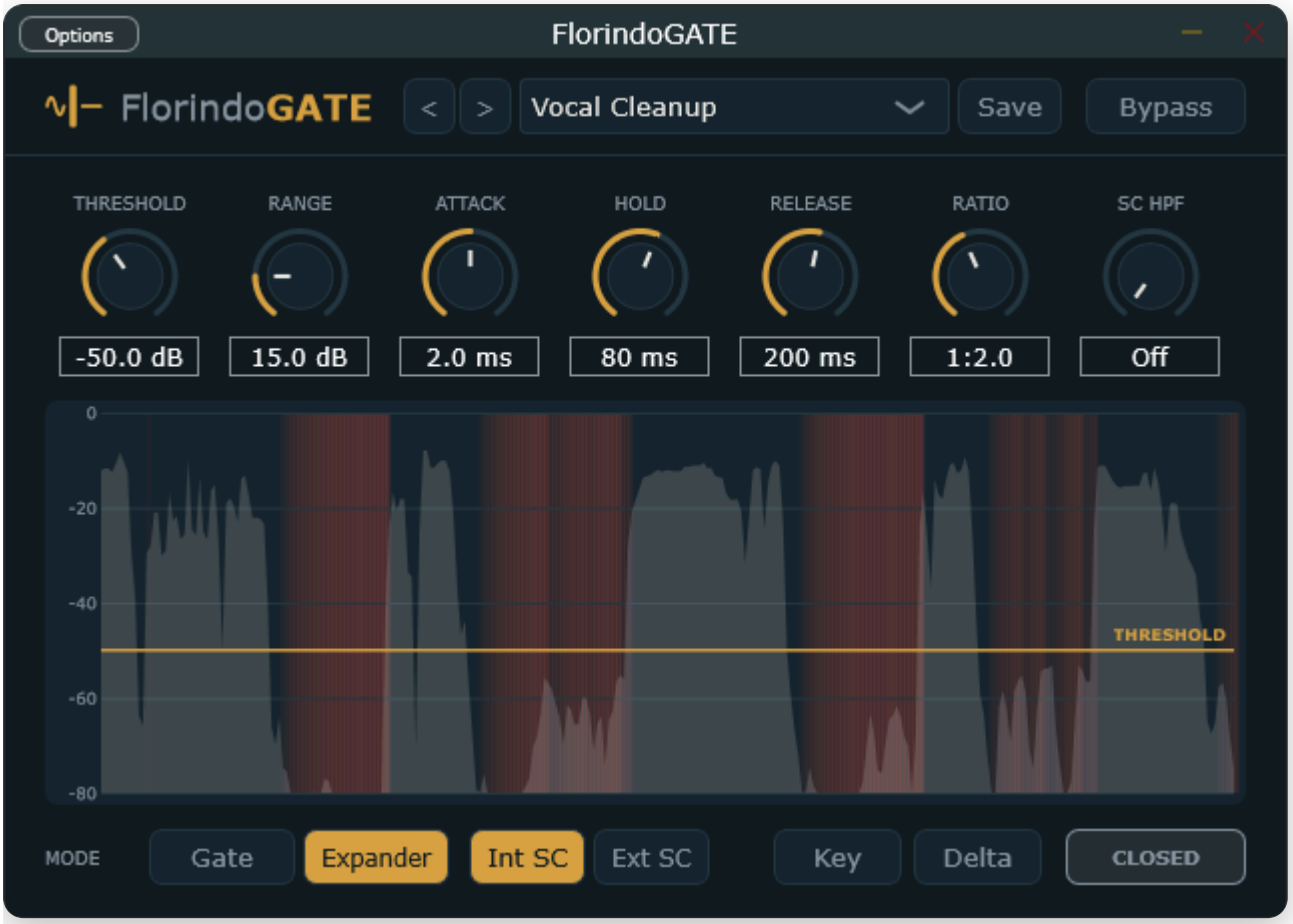
Frequency	monitored area, also adjustable on the spectrum.
Threshold / Range / Release	threshold, maximum reduction, recovery.
Mode	Split (only the highs are reduced, natural) or Wideband (the whole signal is reduced).
Detect	Band or High.
Listen / Delta	what the de-esser detects, what it removes.

> **Tip.** Turn on Listen and move the frequency until you only hear the “s” sounds: the setting is then right.

FlorindoGATE

Gate and expander

Cuts or reduces the signal below a threshold: background noise, bleed between phrases, drum ringing.



SETTINGS

Threshold / Range	threshold and attenuation when the gate is closed.
Attack / Hold / Release	opening, hold, closing.
Mode	Gate (hard) or Expander (gradual, with Ratio).
SC HPF	ignores low frequencies for detection.
Int SC / Ext SC	internal or external detection.
Key / Delta	listen to the detection signal, to what is removed.

> **Tip.** On a voice, a Range of 10 to 15 dB in Expander mode cleans up more naturally than a full cut.

FlorindoLIMIT

True Peak limiter

The end-of-chain limiter: it raises loudness while guaranteeing a strict ceiling.



SETTINGS

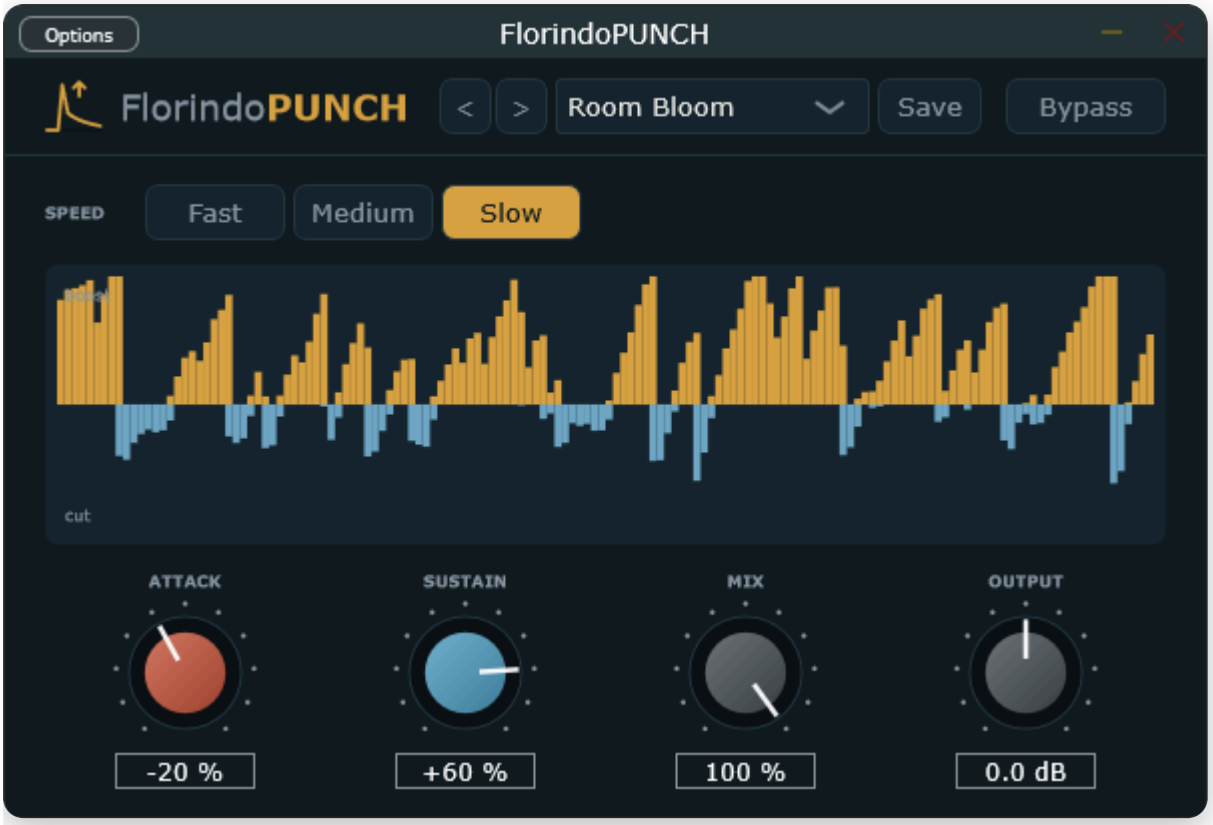
Threshold	drives the signal into the limiter (lower = louder).
Ceiling	absolute ceiling, −1 dB by default.
Release	recovery speed.
True Peak	also watches peaks between samples.
Delta	lets you hear what the limiter removes.

> **Tip.** The limiter looks 2 ms ahead: this latency is reported to the host, which compensates for it automatically.

FlorindoPUNCH

Transient shaper

Hardens or softens the attack of a sound, and lengthens or shortens what follows, with no threshold to set.



SETTINGS

Attack	to the right, attacks hit harder; to the left, they soften.
Sustain	lengthens or shortens the tail of the sound.
Speed	Fast, Medium or Slow depending on the material.
Mix / Output	blend and level.

> **Tip.** On a snare: Attack +50 % and Sustain −20 % for a dry, punchy sound.

FlorindoECHO

Stereo and ping-pong delay

A warm echo that can sync to tempo, with filters and saturation in the feedback loop.



SETTINGS

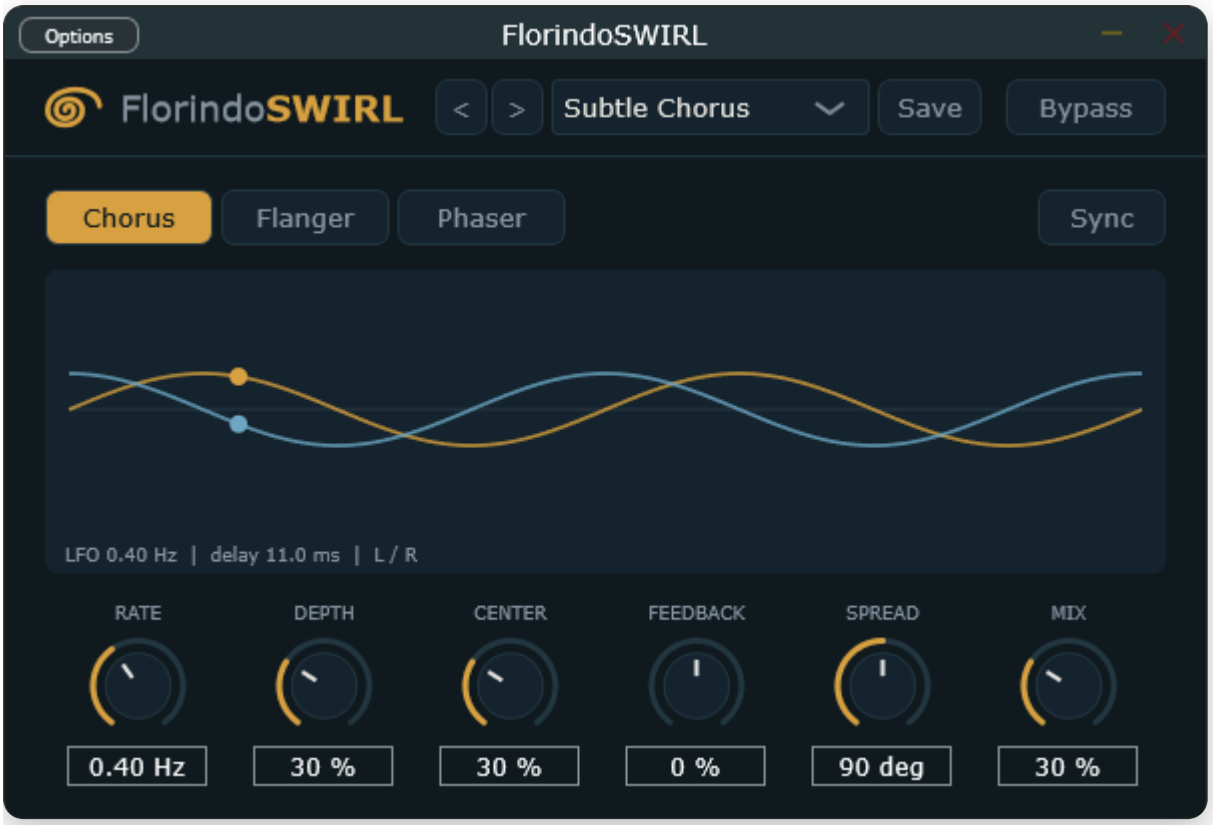
Time / Division	free time, or a rhythmic division of the tempo with Sync.
Feedback / Mix	repeats and blend.
Low Cut / High Cut	filter each repeat a little more.
Saturation / Wobble	warmth and gentle tape-style modulation.
Mode	Stereo or Ping-Pong.

> **Tip.** A 1/8D (dotted eighth) time with 30 % Feedback is a classic on guitars and vocals.

FlorindoSWIRL

Chorus, flanger, phaser

Three modulation effects in one, with a free or tempo-synced LFO.



SETTINGS

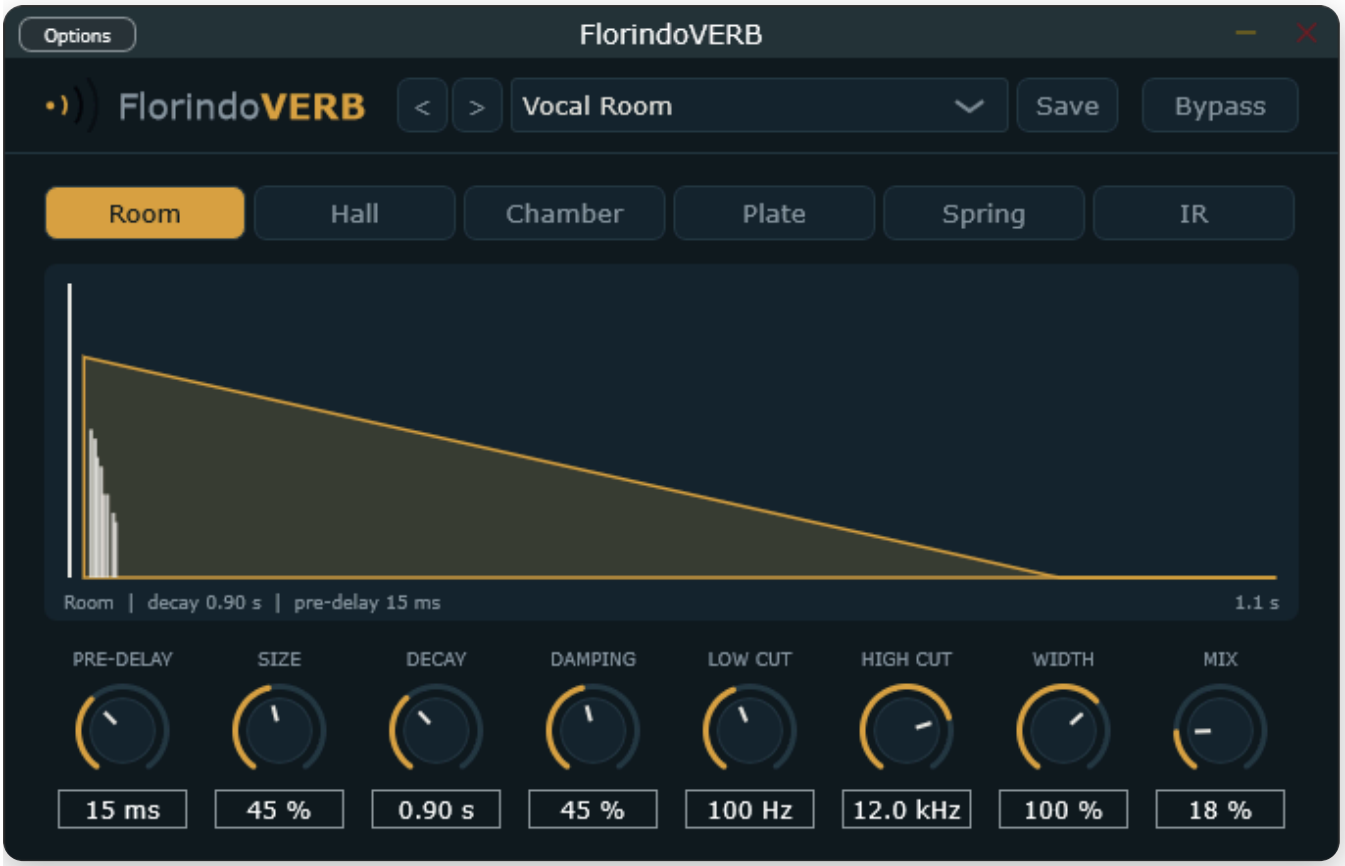
Mode	Chorus, Flanger or Phaser.
Rate / Sync	LFO speed.
Depth / Center	amount and working range.
Feedback	positive or negative feedback.
Spread	offset between left and right.
Mix	effect blend.

> **Tip.** A slow chorus with plenty of Spread beautifully widens a clean guitar or an electric piano.

FlorindoVERB

Reverb and convolution

Five algorithmic reverb types, plus a convolution mode using your own IR files.



SETTINGS

Type	Room, Hall, Chamber, Plate, Spring or IR.
Pre-Delay / Size / Decay	delay before the reverb, size and length.
Damping	high-frequency damping in the tail.
Low Cut / High Cut / Width / Mix	colour, width, blend.
Load IR	in IR mode: loads a WAV, AIFF or FLAC file.

> **Tip.** Cutting the lows of the reverb (Low Cut around 200 Hz) keeps the mix clear.

FlorindoPITCH

Shifts the pitch by semitones, without changing the duration.



SETTINGS

Semitones / **−** / **+**	from −12 to +12 semitones, with the interval name.
Fine	fine adjustment in cents.
Mix / **Output**	blend and level.
Quality	Tight (percussive), Balanced, Smooth (vocals, pads).

> **Tip.** Fine at +12 cents and Mix at 40 % thicken a voice or a guitar.

FlorindoDRIVE

Saturation and soft clipper

Adds grit, warmth or aggression, from tape saturation to hard clipping.



SETTINGS

Type	Tape, Tube, Soft Clip or Hard Clip.
Drive / Tone	amount, darker or brighter colour.
Mix	parallel saturation.
Auto Gain	keeps the level constant so you can judge the texture.
Output	output level.

> **Tip.** Hard Clip with little Drive, at the end of the chain, controls drum peaks better than a limiter.

FlorindoTHUMP

Bass preamp

The bass preamp: grit in the mids, with a round, solid low end thanks to Blend and Low Clean.



SETTINGS

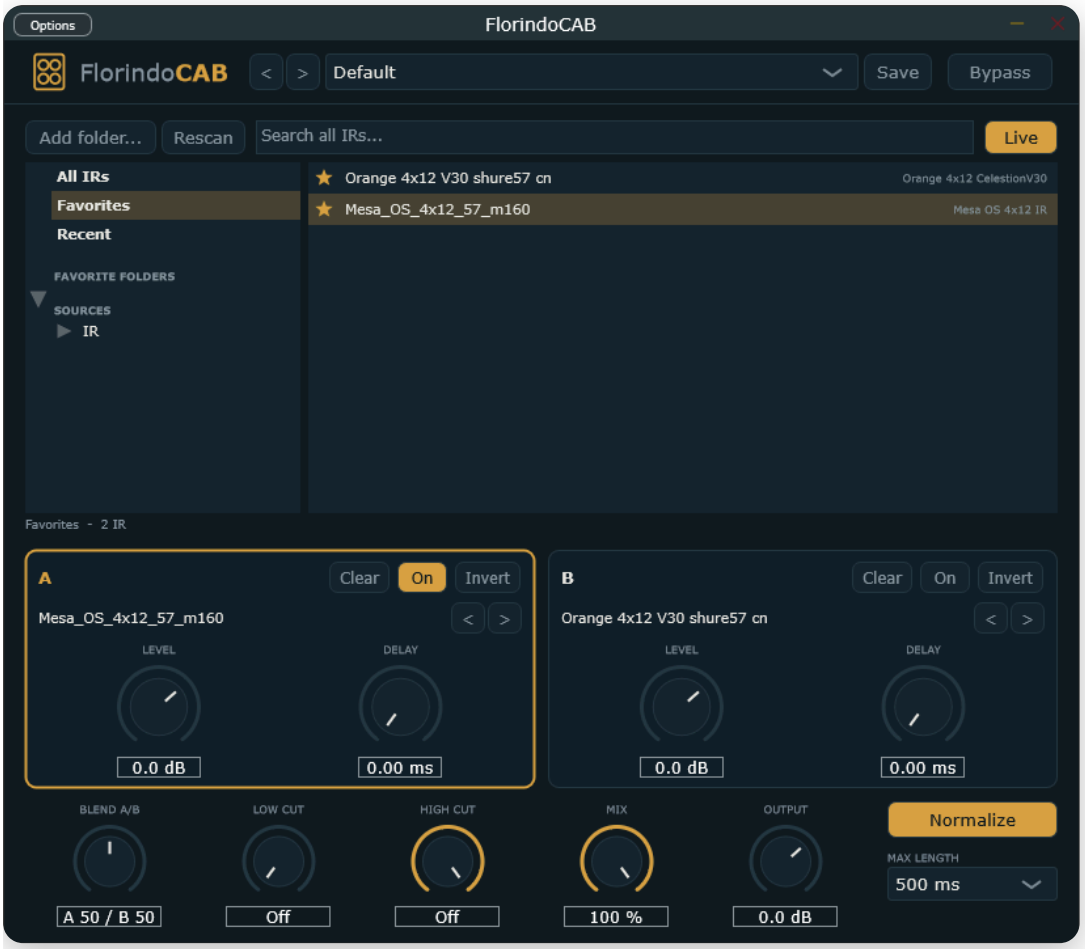
Drive / Character	amount and type: Clean, Grit, Fuzz.
Bass / Mid (+ Mid Freq) / Treble / Presence	tone stack.
Low Clean	lows below 80, 150 or 250 Hz stay clean.
Blend	mix with the direct (DI) signal.
Speaker	built-in speaker simulation. Turn it off if you use CAB right after.

> **Tip.** Start with the Warm Growl preset, then adjust Blend to bring back the bass attack.

FlorindoCAB

A/B cabinet IR loader

Loads cabinet IRs (guitar, bass) with a full library, favourites and instant auditioning.



SETTINGS

Add folder	adds an IR folder (including subfolders).
Navigation	All IRs, Favorites, Recent, favourite folders, folder tree.
Favourites	star or F key for an IR, right-click for a folder.
Live	selecting an IR loads it immediately (arrow keys work too).
A / B	two slots with Level, Delay (phase alignment), On, Invert, Clear.
Blend / Low Cut / High Cut / Normalize / Max Length	global settings.

> **Tip.** Click slot A, turn on Live, then step through the list with the arrow keys while playing: comparison is instant.

INFORMATION

Licence and contact

FlorindoAudioSuite is free, for personal and professional use. The full terms are in the LICENCE.txt file supplied with the plug-ins, and third-party components (JUCE, Steinberg VST3 SDK) are detailed in THIRD-PARTY-NOTICES.txt.

VST is a registered trademark of Steinberg Media Technologies GmbH.

Updates, other free tools and contact: **florindolab.com**

The illustrations in this guide are screenshots of the standalone applications, with example settings.